Syllabus for CS666: Biomedical Signal and Image Processing

Computer Science for Biomedical Diagnostics and Research!

This course introduces important signal and image processing methods for biomedical diagnostics and research. You will learn hands-on how to reconstruct, visualize, and analyze datasets from different modalities such as electrocardiography (ECG), electroencephalography and magnetoencephalography (EEG/MEG), ultrasound (US), X-ray, electron and light microscopy (EM/LM), computerized tomography (CT), structural and functional magnetic resonance imaging (MRI/fMRI), as well as single photon emission computed tomography and positron emission tomography (SPECT/PET). Course discussions and assignments include the fundamentals of digital signal processing, filtering and denoising, Fourier transformations, pattern recognition, and state-of-the-art registration and segmentation pipelines. After completion, you will have the $kills to work at hospitals, life science institutions, and biotech companies!

We will examine, learn, and use:

- Real-world Data acquired using ECG, EEG/MEG, US, X-ray, EM/LM, CT, MRI/fMRI, and SPECT/PET
- Popular Medical Applications such as 3D Slicer, The ChRIS Project, SliceDrop.com, and many others..
- Medical Signal and Image Analysis with Python (Pandas, OpenCV, SciKit-Image, Mahotas..)
- Applied Deep Learning for Medical Imaging with Keras/TensorFlow
- Recent Research from the International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)
- Github / git and Overleaf / LaTeX

Teaching Staff

Instructor: Daniel Haehn
Teaching Assistants/Graders: TBA
Contact: `@staff` on Discord

Lectures

Mondays and Wednesdays
1:00pm-12:15pm
Wheatley-Peters W02-0126

Office Hours

Mondays and Wednesdays
11:30pm-1:00pm and by request
McCormack M03-2120, please use `https://calendly.com/haehn/` to reserve a slot.

Canvas Access

Please use Canvas to access lecture videos, slides, and all other materials.
Login at `https://umassboston.instructure.com/`.

Discord Access

The teaching staff is available via Discord. The invite link will be shared in class.

Questions and Concerns

Please direct questions and concerns of any kind (now and during the semester) to the teaching staff in person or via Discord @staff or in the #help channel.

Course Structure

26 Lectures
7 Assignments (30% of final grade, plus 10% bonus)
10 Quizzes (20% of final grade, take-home through Canvas)
Journal Club (30% of final grade)
Participation (in-class, in-office, and as part of Discord discussions, 20% of final grade)

We will have **multiple guest lectures** from experienced researchers.

Final Grade

The weighted scores from above will result in a final grade as follows:

$\geq$ 90 = A	69-66 = C
89-86 = A-	65-62 = C-
85-82 = B+	61-58 = D+
81-78 = B	57-54 = D
77-74 = B-	53-50 = D-
73-70 = C+	below 50 = F

Interactive Lectures

Lectures will include interactive components. If you do not have a laptop or reliable internet, please contact the teaching staff via Discord.

Assignments

Assignments include research questions and coding challenges. All assignments require a written report. You will use a standard git workflow to submit your work. Instructions and templates are available in the Canvas system. **Assignments are due Mondays at 11:59pm. No late submissions.**

Quizzes

Quizzes include multiple-choice and free-text questions. They are take-home quizzes and available in the Canvas system. **Quizzes are due Fridays at 11:59pm. No late submissions.**

Journal Club

We will read classic and recent papers during the weekly journal club sessions. You will present 1-2 papers per semester during these sessions followed by class discussions.

Participation

In-class participation and Discord activity count towards your grade. If class attendance drops below 50%, surprise tests may happen during lectures.

Collaboration Policy

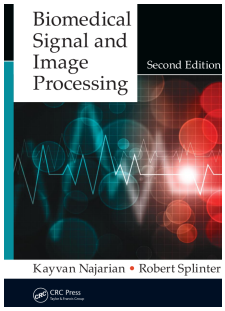
You are allowed and encouraged to collaborate with anybody. However, please make sure to give proper credit. For instance, if your friend helps you with your report or you copied code from another source, you must acknowledge their name in your code and the project documentation. **ChatGPT or any other AI tool requires credit and the student is responsible to double-check the content!**

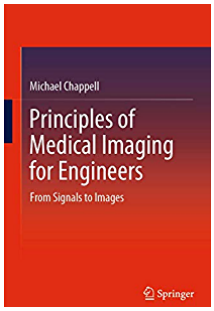
Open Source License and Proprietary Code

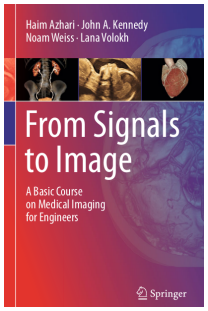
The course material is publicly available under the MIT license (`https://opensource.org/licenses/MIT`). This includes provided code. You are free to adopt a different license for the code you write in the course.

Readings

The course material is based on the following books:







- Biomedical Signal and Image Processing by K. Najarian and R. Splinter
- Principles of Medical Imaging for Engineers by M. Chappell
- From Signals to Image by H. Azhari, J.A. Kennedy, N. Weiss, and L. Volokh

Limited copies of all books are available through the teaching staff. While the books are great, **you do not need to purchase them**—the most up-to-date information is available online.

Disability Accommodations

If you have a disability and feel you will need accommodation to complete course requirements, please contact the Ross Center for Disability Services at 617.287.7430.

Other Policies

We follow the Academic Policies of the Office of the Registrar.
See `https://www.umb.edu/registrar/academic_policies` or contact `staff@cs410.net` for questions.

Timeline

Date			Lecture	Due at 11:59pm
01/26/2026	M		No Class (Snow Day)	
01/28/2026	W	01	Introduction	
			Journal Club Prep: Intro to Research	
02/02/2026	M	02	Signals and Images I	
02/04/2026	W	03	Signals and Images II	
			Journal Club Prep: How to read a paper	Quiz 1
02/09/2026	M	04	Electrocardiography	
02/11/2026	W	05	Data Wrangling	
			Journal Club Prep: How to read a paper II	Quiz 2
02/16/2026	M		No Class (President's Day)	
02/18/2026	W	06	TBA	
				Quiz 3
02/23/2026	M	07	No Class (Snow Day)	Assignment 1
02/25/2026	W	08	Segmentation and Registration	
03/02/2026	M	09	Visualization	Assignment 2
03/04/2026	W	10	Processing Frameworks	
				Quiz 4
03/09/2026	M	11	Ultrasound	Assignment 3
03/11/2026	W	12	NeuroDesk	Quiz 5
03/16/2026	M		No Class (Spring Break)	
03/18/2026	M		No Class (Spring Break)	
03/23/2026	M	13	X-Ray	Journal Club 1
03/25/2026	W	14	Microscopy	Journal Club 2, Quiz 6
03/30/2026	M	15	Guest Lecture: Kai Kang (Etiometry)	Assignment 4
04/01/2026	W	16	CT	Journal Club 3, Quiz 7
04/06/2026	M	17	MRI/fMRI	Journal Club 4, Assignment 5
04/08/2026	W	18	TBA	Journal Club 5, Quiz 8
04/13/2026	M	19	Applied Deep Learning	Journal Club 6
04/15/2026	W	20	Applied Deep Learning II	Quiz 9
04/20/2026	M		No Class (Patriots Day)	
04/22/2026	W		No Class (Assignment 6)	
04/27/2026	M	21	SPECT/PET	Journal Club 7, Assignment 6
04/29/2026	W	22	Guest Lecture: Michal Depa	Quiz 10
05/04/2026	M	23	Biometrics	Journal Club 8
05/06/2026	W	24	TBA	Journal Club 9
05/11/2026	M	25	Future and Outlook	Journal Club 10, Assignment 7
05/13/2026	W	26	Final Recap	